

Impact of urban morphology on outdoor air temperature and microclimate optimization strategy base on Pareto optimality in Northeast China

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ABSTRACT

The urban microclimate is affected by the urban morphology parameters. The objective of this study was to investigate the correlation between air temperature and urban morphology parameters in a cold-climate city in China. Field measurements were performed to record the microclimate parameters at 27 points in a 10 km² urban area in Harbin, China. Multilinear regression models were created to investigate the effects of the urban morphology parameters on the outdoor air temperature. Several interesting findings were obtained. First, the degree of influence of urban morphology on temperature was time-dependent and spatial-dependent, and the largest influence was observed in a 100 m radius around the point of interest. Second, an increase in the green plot ratio of 0.5 decreased (increased) the daytime average temperature by 0.8 °C (1.3 °C) in the summer (winter). For the same street width, a 10 m increase in the building height decreased the daytime temperature by 0.06 °C, and for the same building height, a 10 m increase in the street width increased the daytime temperature by 0.17 °C in summer. The daytime average temperature was significantly affected by the number of building floors; however, when the building had more than 20 floors, the temperature did not change significantly with the building height. In addition, a multi-objective optimization framework based on Pareto optimality was used to ascertain the optimal urban morphology layout considering the urban microclimate. The optimal wind and thermal environments were achieved at building densities ranging from 20% to 30% and building plot ratios ranging from 3 to 5.

1. Introduction

The urban heat island (UHI) effect is increasing due to increasing urbanization. The urban microclimate is the result of highly dynamic and complex processes that depend on the macroclimate and urban morphology [1]. The microclimate affects human comfort and the utilization of outdoor spaces [2]. Therefore, research on the microclimate of specific cities or sites is required to improve our understanding of the parameters that affect the urban microclimate.

The impact of urban morphology on the outdoor microclimate has received significant research interests in recent decades. The temperature inside buildings depends on the proportions of building and vegetation, the building height, wall area, road proportion, and other elements, as well as the regional climate [3,4]. The urban morphology affects the microclimate to varying degrees [5]. Field research by Jusuf and Wong [6] showed that within a 50-m radius, the green plot ratio (GnPR), total leaf area of trees, and the percentage of the green area

were critical parameters with significant influences on the minimum temperature and average temperature. The sky view factor (SVF) had a significant influence on the maximum temperature. Field research in Beijing by Yan [7] indicated that urban morphology had the most significant impact on the temperature within a 75-m radius, and this range was the same as that in Tokyo [8]. The urban geometry and the thermal properties of urban surfaces were also important parameters influencing the microclimate [9]. Land cover features, such as the ratio of building to vegetated areas, affect the microclimate. The SVF, which depends on the site geometry, is another crucial factor affecting the mean radiant temperature [10]. However, some studies have shown that the SVF had a negligible impact on air temperature [11,12]. The air temperature is correlated with the green ratio and the building ratio and depends on the time of year (day) and the weather [13–15]. The parameters that affect the outdoor microclimate and the radius of influence in cold cities such as Harbin remain unclear.

Empirical models that describe the relationship between urban morphology parameters and microclimate parameters based on field

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Nomenclature

<i>BDG</i>	Percentage of building area over a surface area
<i>HBDG</i>	Ratio of the height of building to the percentage of building area
<i>GnPR</i>	Green plot ratio
<i>LAI</i>	Leaf area index
<i>PAVE</i>	Percentage of pavement area (%)
<i>WALL</i>	Total wall surface area over a 50 m-radius surface area (m ²)
<i>SVF</i>	Sky view factor
<i>RH</i>	relative humidity (%)
<i>WIND</i>	average wind speed (m/s)
<i>T_{avg}</i>	Daily average temperature (°C)
<i>T_{max}</i>	Daily maximum temperature (°C)
<i>T_{min}</i>	Daily minimum temperature (°C)
<i>Ref T_{avg}</i>	Daily average temperature at a reference point (°C)
<i>Ref T_{max}</i>	Daily maximum temperature at a reference point (°C)
<i>Ref T_{min}</i>	Daily minimum temperature at a reference point (°C)
<i>SOLAR</i>	Daily solar radiation at a reference point (W/m ²)

measurements are commonly used to assess the influence mechanism of urban morphology on the microclimate. Kruger and Givoni [16] developed equations to correlate the maximum, minimum, and average air temperatures with the water area, built area, open area, paved area, and green area in Curitiba. The equations were used to predict the temperature of areas with the same urban morphology. Tong [17] applied the same method in Tianjin, China and found that an increase in the number of building levels and reduction in pavement width lowered the T_{max} and $T_{avg-day}$ and increase the T_{avg} , T_{min} , $T_{avg-night}$. Moreover, an increase in the $GnPR$ of 0.5 lowered the T_{min} and $T_{avg-night}$ by 0.7 °C and 0.5 °C respectively.

Outdoor microclimate optimization should consider the influence mechanism of urban morphology on each climate factor. The optimization of a single goal may adversely affect other goals. Therefore, the comprehensive optimization of multiple microclimatic factors is of practical significance to improve the urban environment. We use the strength Pareto evolutionary algorithm (SPEA-2) [18] to develop design-based recommendations. Schaffer [19] proposed the multi-objective vector evaluated genetic algorithm (VEGA) algorithm in 1985. Subsequently, Goldberg and Pareto [20] proposed the Pareto frontier theory to calculate the optimal solution set. SPEA-2 is the revision of a previously developed algorithm and provides improved accuracy of the optimal solution [21]. Previous researchers have used genetic algorithms for the optimization of the energy balance of the building environment and traffic optimization. However, few studies have considered wind speed, sunshine duration, and temperature as optimization objectives to develop urban design-based recommendations.

The objective of this study is to investigate the effects of urban morphology on temperature and use the SPEA-2 algorithm to develop an urban morphology optimization strategy to improve the thermal environment. Meteorological data were obtained in the field to investigate the spatial and temporal microclimate parameters in Harbin. The influence of the urban morphology parameters on the air temperature and the radius of influence of the urban morphology parameters were analyzed. A multi-objective optimization framework was used to determine the optimal urban morphology considering the urban microclimate, and design-based recommendations were presented for the early stage of urban design.

2. Methodology

2.1. Study area

Harbin (125°42' - 130°44'E longitude, 10°04' - 46°40'N latitude) is the capital of Heilongjiang province and the largest city in northeastern China. The area has a temperature continental monsoon climate with long winters and short summers [22]. The annual average precipitation is 569.1 mm, and the snow period is from November to January. The T_{avg} is -19 °C in winter and 23 °C in summer. The study area is 10.1 km², with a perimeter of 6.5 km and includes pedestrian streets, plazas, and small parks. The urban area has an open block layout. The study area is characterized by high building density, moderate vegetation cover, multi-story and high-rise buildings, and diverse block types; the morphological characteristics and functional types are those of open blocks. The layout of the study area is shown in Fig. 1.

2.2. Microclimate parameter measurements

The outdoor microclimate was assessed using 27 measuring points arranged in a grid with a distance of 100–200 m, as shown in Fig. 1. Most of the block morphology in the area was covered by the measuring points. The microclimate parameters that were measured included air temperature, relative humidity, and wind speed. The microclimate station is shown in Fig. 2, and the sensor precision is listed in Table 1. Measurements were obtained automatically and synchronously at each location every 5 min. The sensors were wrapped in aluminum foil, leaving a gap in the middle to minimize the influence of solar radiation on the temperature. The weather stations were placed at the height of 1.5 m on a tripod to obtain data at the height of pedestrians.

The focus of this study was to determine the impact of urban morphology on air temperature during summer and winter; therefore, fairly clear, calm, and sunny weather conditions were preferred so that the data represented typical winter or summer weather conditions [23]. According to the Harbin observatory, the T_{avg} should be above 25 °C in summer and below -7 °C in winter to represent typical summer and winter meteorological conditions. The average wind speed in an urban area should be lower than 3 m/s at the height of a pedestrian. Therefore, the days chosen for the analysis in this study had to match the following conditions:

- The maximum solar radiation had to be higher than 800 W/m² in summer and lower than 900 W/m² in winter;
- The hourly solar radiation and air temperature profiles had to be normally distributed;
- The daily air temperature had to be higher than 25 °C in summer and lower than -7 °C in winter;
- The daily average wind speed had to be less than 3 m/s, and there should be no rain or snow.

The measuring points were located in open areas of the block. Since several researchers were required to monitor the sensors, continuous daily monitoring was not possible. Therefore, we selected 40 d during Dec. 2017 to Aug. 2018 that met the conditions, and the plan was confirmed based on the weather forecast. Some modifications were made due to wrong weather forecasts and instrument failure; we eventually selected 35 d and randomly divided them into two groups used for model development (30 d) and validation (5 d), respectively (Table 2). The measurements were conducted from 0:00 to 24:00 continuously on each day.

2.3. Urban morphology parameters measurement and computation

Geographic Information System (GIS) mapping is an effective method to link the urban morphology parameters with the local microclimate [24]. Urban planning is influenced by history and culture,

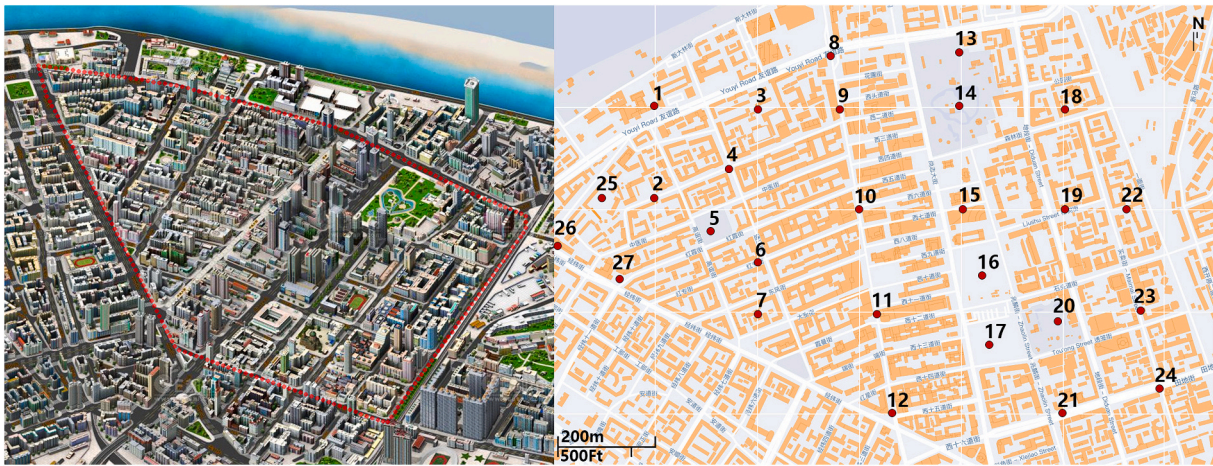


Fig. 1. Layout of the study area and measurement points set

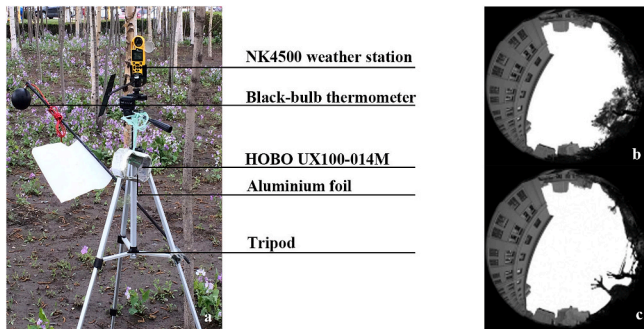


Fig. 2. (a) Example of installed weather station, and fish-eye pictures taken in (b) summer and (c) winter

Table 1
Technical characteristics of measurement instruments.

Temperature/RH	WBGT-2006
	Global temperature (T_g)
	–40 °C–70 °C, ± 0.18 °C
	HOBO UX100-014M
	Temperature range/accuracy
	–40 °C–70 °C, ± 0.2 °C
	RH measurement range/accuracy
	0%–100%, $\pm 2.5\%$
Wind speed/direction	NK4500 weather station
	Wind direction range
	2-Axis ultrasonic wind sensor
	Wind speed range/accuracy
	0–45 m/s (0–100 mph) ± 1.1 m/s(2.4mph)
Sky view factor (SVF)	Canon 5D2 camera with fish eye lens

Table 2
Selected days for model development and validation.

30 days for model development		5 days for model validation	
Dec 2017	10, 18, 20	Dec 2017	17
Jan 2018	11, 13, 15, 25	Feb 2018	15
Feb 2018	2, 8, 9	Apr 2018	14
Mar 2018	5, 7, 8	Jun 2018	18
Apr 2018	2, 5, 6	Aug 2018	20
May 2018	20, 26, 27		
Jun 2018	22, 23, 25, 26		
Jul 2018	5, 8, 16		
Aug 2018	5, 6, 10, 18		

and urban morphology has individual characteristics [25]. In reference to international studies, the following four criteria should be met in the selection of urban morphology parameters:

- 1 The parameters should significantly affect the microclimate.
- 2 The parameters should be easy to extract and calculate.
- 3 The parameters affect the design.
- 4 Parameter redundancy should be avoided.

Several parameters were chosen in this study. The *GnPR* affects the regional temperature by reducing heat loss through evaporation. The *SVF* affects solar radiation. The *HBDG* affects the solar heat gain resulting from solar radiation. The *WALL* affects the indoor and outdoor heat exchange, and the *PAVE* affects ground surface heat exchange. A quantitative GIS analysis of the parameters was conducted to determine the influence of the urban morphology parameters on the temperature.

As shown in Fig. 3a, the measurement points are located in different land cover classes, resulting in differences in the temperature range. Therefore, the morphology of the points was quantified after establishing buffer zones with radii of 20 m, 50 m, 100 m, and 150 m, as shown in Fig. 3b.

The values of *PAVE*, *HBDG*, and *WALL* were extracted from the GIS data. We used an 8-mm circular fisheye lens to obtain images that were imported into the Rayman model to obtain the *SVF* values [26]. For the *GnPR* calculation, we used the following equation [27]:

$$GnPR = \sum (n_m A_m \times LAI_m) / \text{Site Area} \quad (1)$$

where LAI_m is leaf area index of species m , A_m is canopy area of species m , n is the number of plants of species m on the site. The morphology information of each test point was shown in Table 3.

2.4. Regression analysis and model development

Multiple regression analysis was conducted using the following steps:

Step 1. We obtained the microclimate data at the measurement points, including air temperature, wind speed (*WIND*), and relative humidity (*RH*). We calculated the morphology parameters of the measurement points in the 20 m, 50 m, and 100 m buffer zone using the GIS data.

Step 2. We established regression models to describe the relationship between the air temperature and the morphology parameters. The air temperature was the dependent variable, and the reference microclimate data (*Ref T*, *SOLAR*) were obtained from the observatory, the measured data (*WIND*, *RH*), and the morphology parameters were the



Fig. 3. (a) Morphology variation in different radius (b) Block layout with buffer zone

Table 3

Description of measuring points (100 m).

No	SVF		GnPR		BDG (%)	HBDG (m)	PAVE(%)	WALL (m ²)	Description of measurement points
	Winter	Summer	Winter	Summer					
1	0.58	0.52	0.34	0.55	15	2.7	30	3035.34	Broad street with avenue tree, multi-storey buildings
2	0.50	0.39	0.198	0.32	35	6.3	33	7082.46	Broad street with avenue tree, multi-storey buildings
3	0.80	0.40	0.427	0.69	21	7.56	10	8498.952	High-rise buildings, shade tree
4	0.61	0.35	0.341	0.55	12	2.52	33	2832.984	Broad street with avenue tree, multi-storey buildings
5	0.78	0.69	0.372	0.6	0	0	94	0	Open area with trees and lawn
6	0.49	0.36	0.136	0.22	20	3.6	56	4047.12	Broad street with avenue tree, multi-storey buildings
7	0.52	0.43	0.341	0.55	35	17.5	10	19673.5	High-rise buildings, shade tree
8	0.65	0.60	0.148	0.24	12	1.44	64	1618.848	Open area with few trees
9	0.92	0.48	0.18	0.30	28	4.2	42	4721.64	Pedestrian street with avenue tree, multi-storey buildings
10	0.40	0.36	0.09	0.15	50	7.5	35	8431.5	Pedestrian street with avenue tree, multi-storey buildings
11	0.32	0.20	0.13	0.18	40	6	42	6745.2	Pedestrian street with avenue tree, high-rise buildings buildings
12	0.26	0.24	0.074	0.12	36	5.4	52	6070.68	Pedestrian street with avenue tree, high-rise buildings buildings
13	0.22	0.19	0.48	0.77	18	0.54	5	607.064	Park's perimeter road, tree cover
14	0.37	0.09	0.496	0.80	8	0.24	12	269.808	Park's perimeter road, tree cover
15	0.31	0.31	0.078	0.12	77	27.72	11	31162.824	High-rise buildings, without tree
16	0.66	0.66	0.31	0.5	68	28.56	17	32107.152	High-rise buildings, without tree
17	0.62	0.62	0.17	0.25	30	5.4	45	6070.68	Enclosed building without tree
18	0.20	0.18	0.111	0.18	42	10.08	40	11331.936	Broad street without tree, multi-storey buildings
19	0.55	0.45	0.12	0.20	46	8.28	34	9308.376	Broad street with avenue tree, multi-storey buildings
20	0.72	0.72	0.49	0.8	7	0.42	85	472.164	Open area with few tree
21	0.58	0.34	0.496	0.8	36	6.48	56	7284.816	Broad street with avenue tree, High-rise buildings
22	0.50	0.50	0.31	0.5	48	8.64	47	9713.088	Broad street with avenue tree, multi-storey buildings
23	0.53	0.42	0.111	0.18	59	10.62	25	11939.004	Broad street without tree, multi-storey buildings
24	0.42	0.37	0.191	0.26	15	2.7	59	3035.34	Broad street with avenue tree, High-rise buildings
25	0.44	0.33	0.094	0.12	32	5.76	56	6475.392	Enclosed multi-storey building without tree
26	0.41	0.37	0.09	0.14	38	4.56	48	4104	Open area without trees
27	0.40	0.40	0.22	0.34	21	3.78	45	3024	Open area with few trees

independent variables. The regression model is defined as:

$$T = a + b_1 \text{Ref } T + b_2 \text{SOLAR} + b_3 \text{WIND} + b_4 \text{RH} + b_5 \text{HBDG} + b_6 \text{GnPR} + b_7 \text{SVF} + b_8 \text{PAVE} + b_9 \text{WALL} \quad (2)$$

where a is the constant, b_n is regression coefficient, and other variables can refer to Nomenclature part. The model was built based on integrated outdoor temperature, so for dependent variables, T include the T_{avg} , T_{max} , T_{min} , and independent variables, $\text{Ref } T$ was the corresponding reference average, maximum, minimum temperature.

Step 3. The results were calculated in the different buffer zones. We established quantitative relationships for one-to-one correspondences to determine the degree of influence of the factors in the range.

Step 4. We calculated the Pearson correlation coefficient (r), the

coefficient of determination (R^2), and the significance (p -value) of the independent variables. If the R^2 was larger than 0.7 and the p -value denoted by Sig (2-tailed) was smaller than 0.05, the variable was used in the final equation. The strength of the linear relationship was evaluated. If the R^2 between two independent variables was larger than 0.7, the variables were considered to be self-dependent and were excluded from the equations. The R^2 of the equations for the different buffer zone were compared, and the equation with the largest R^2 was used as the final model.

2.5. Multi-objective optimization model

Each urban morphology parameter has a different influence on each microclimate parameter. For example, an increase in the street height to width (H/W) ratio can lower the air temperature and solar radiation and increase the wind speed, resulting in a comfortable thermal environment

but an uncomfortable wind environment. In addition, a morphology that provides a comfortable thermal environment in the summer may result in an uncomfortable environment in the winter. In these cases, a multi-objective optimization using the Pareto optimal solution set is suitable [20].

The evaluation criteria for this solution are as follows: If the output performance is balanced when all optimization objectives are considered, the performance improvement of one optimization objective is greater than or equal to that of another optimization objective. The solution set with a balanced performance is called the non-dominant solution set. The solution set with the best equilibrium performance is called the optimal solution set. The non-dominant solution set distribution is called the Pareto frontier. The ultimate goal of multi-objective optimization is to obtain the Pareto frontier solution set. The mathematical definition of multi-objective optimization is as follows:

$$\min f(x) = [f_1(x), f_2(x), \dots, f_p(x)]^T,$$

$$g_i(x) \geq 0, \quad i \in I,$$

$$h_j(x) = 0, \quad j \in E \quad (3)$$

where for multi-objective programming (MOP), the variable feasible region is S , and the target feasible region is $Z = f(S)$. For a viable solution $x^* \in S$, solution should be $\forall x \in S, f(x^*) \leq f(x)$; therefore, x^* is the optimal solution for multi-objective optimization. If $x \in S$ does not exist, then $f(x) < f(x^*)$, where x^* is called the efficient solution or the Pareto optimal solution of MOP.

After determining the influence of the urban morphology of Harbin on the microclimate, we conducted a sensitivity analysis, simplified the parameters, and used Grasshopper plug-in Ref. [28] of the Rhino modeling software to conduct parametric modeling of the urban morphology. For the microclimate simulation, the Radiation software was used to calculate the solar radiation and the sunshine duration, the Ladybug software was used to calculate the temperature using the proposed equations, and the Butterfly software was used to calculate the wind speed. For the multi-objective optimization, Octopus software was used to calculate the optimal solution set according to the simulation data. All the software modules were integrated using Grasshopper plug-in.

Many urban morphology indicators influence the accuracy of the optimization result, such as *BDG* and *HBDG*. Therefore, the complex urban morphology parameters were simplified into basic morphological variables as building height, building row spacing, column spacing, and

building layout to guide the urban design. We then used the optimal solution set and calculated the *BDG*, *PAVE*, *HBDG*, and other urban design indicators using the basic morphological variables. The optimization logic is shown in Fig. 4.

The basic situation of the urban block layout was investigated in Harbin. The constraints for the model establishment and the simplification of the urban morphology parameters were as follows based on the field survey.

Site area: the basic block unit is 400 m × 400 m, the road width is 15 m–25 m, and the building height is 0–150 m, where 0 indicates an open space. The block gap rate is an essential factor determining the thermal environment and wind environment. The row spacing and column spacing are the only factors that determine the gap rate. According to the field survey, we simplified the conditions at the simulation site as 3–5 columns and 3–5 rows. The column separation distance was >1.8 times the building height and <50 m, according to the city planning department needs. The minimum building area should be > 750 m² and <5000 m². The mathematical expressions are as follows:

$$\begin{aligned} 0 &\leq H \leq 150 \\ 50 &\geq D_1 \geq 1.8H \\ 50 &\geq D_2 \geq 6 \\ \left\{ \begin{aligned} L &= \frac{400 - (a-1)D_1}{a} \quad (3 \leq a \leq 5) \\ W &= \frac{400 - (b-1)D_2}{b} \quad (3 \leq b \leq 5) \end{aligned} \right. \quad (4) \\ 5000 &\geq L^*W \geq 750 \end{aligned}$$

where H is the building height, D_1 is the building row spacing, D_2 is the column spacing, L is the building length, W is the building width, a is the number of rows, and b is the number of columns. The Grasshopper software can generate all possible layouts of the buildings using these equations.

A previous study on human thermal comfort in Harbin indicated that the summer outdoor microclimate is comfortable for tourists. However, in the winter, solar radiation and wind speed are the primary factors causing discomfort for tourists [29]. Therefore, in this study, the optimization objectives were to increase the sunshine duration in the winter, increase the heat gain of the site, and reduce the overall wind speed.

Optimization target for solar radiation: the objective was to increase the average sunshine hours. We selected the average sunshine hours (h) on the south facade of the building on a cold day at 1.5 m. The

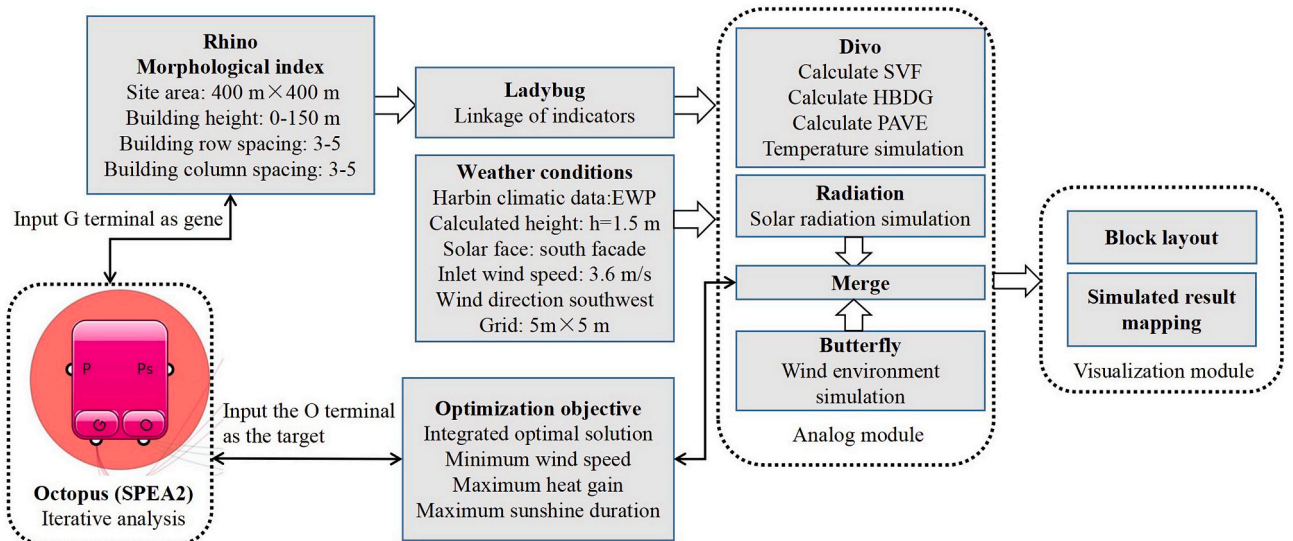


Fig. 4. Optimization logic

optimization objective function is $\alpha = 1/h$.

Optimization target for the thermal environment: the objective was to increase the maximum heat gain of the site. The average solar radiation of the site in winter is I , and the optimization objective function is $\gamma = I$.

Optimization target for the wind environment: the objective was to reduce the average wind speed (V_{avg}), and the optimization objective function is $\beta = 1/V_{avg}$.

3. Results and analysis

3.1. Influence of temporal and spatial variation on the microclimate

Fig. 5 shows the temperature mapping at the different measuring points during the daytime (14:00) and nighttime (2:00) in summer. Fig. 6 shows the temperature mapping during the daytime (14:00) and nighttime (2:00) in winter.

a. Daytime temperature distribution in the summer. As Fig. 5a shows, the temperatures were significantly different at the different locations, even in a small area, due to the differences in *HBDG*, *SVF*, *GnPR*, and *PAVE* at each measuring point. The highest temperature difference was 3.88 °C. The highest temperature of 29.19 °C occurred at measuring points 5. The lowest temperature was 25.31 °C at measuring points 13 and 14. The temperatures at measuring points 5, 16, and 17 in the square were significantly higher than those of the other measuring points during the day. The reason is the absence of shelter in the square, resulting in heat absorption. The lowest temperature occurred in the park, where the dense vegetation reduced the temperature. It is worth noting that the temperature of the measuring point in the central street was significantly lower than that in the surrounding area; this result was attributed to the lack of vehicles in the pedestrian area and the dense landscape trees on both sides of the streets that block the sunshine, resulting in relatively low temperatures.

b. Nighttime temperature distribution in the summer. As Fig. 5b shows, the maximum temperature difference was 2.37 °C; the highest temperatures occurred at points 22 and 23 (19.19 °C), and the lowest temperature was observed at point 13 (16.82 °C). The temperatures at measuring points 5, 16, and 17 in the square were significantly lower than the points in the surrounding area, which was due to the high *SVF* and rapid heat dissipation at night. The temperatures of the measuring points near the park were significantly lower than those of the other areas, showing heat diffusion. A “cool island effect” occurred in the dense trees, which reduced the surrounding temperature.

a. Daytime temperature distribution in the winter. As Fig. 6a shows, the maximum difference in the daytime temperature (1.5 °C) was lower in winter than in summer. The highest temperature (−19.2 °C) was observed at point 3, and the lowest temperature (−20.7 °C) occurred at point 14. The temperatures at points 5, 16, and 20 were significantly

higher than those of the other points in the surrounding area. However, the temperatures of points 13 and 14 in the park were lower than those of the other areas, although the leaf area index (*LAI*) index is low in the winter. The low temperatures in the area are attributed to the lack of indirect radiation from the buildings, the tree branches that block solar radiation, and the snow covering large areas.

b. Nighttime temperature distribution in the winter. As Fig. 6b shows, the maximum nighttime temperature difference during winter was 5.1 °C. The highest temperature was −22.8 °C in the area near point 18, and the lowest temperature was −28 °C at point 9. The trends of the daytime and nighttime temperatures exhibited opposite trends. The reason is that in the winter, there is no green vegetation, and solar radiation is the main factor influencing the air temperature. The *SVF* affects the outdoor heat gain, and the *GnPR* and *HBDG* affect the heat loss in the region, resulting in the different trends of the daytime and nighttime temperatures.

Different fluctuations in the temperature were observed in the two seasons. As Fig. 7 shows, the degree of influence of the urban morphology on the temperature depended on the season. In the summer, the temperature differences were in the range of 2.6–4.68 °C during the day and 1.75–2.37 °C during the night. In the winter, the range was 2.45–3.6 °C during the day and 3.33–4.6 °C during the night. The ranking of the temperature range in the two seasons, which reflects the degree of influence of urban morphology on the temperature, was summer daytime temperature > winter nighttime temperature ≈ winter daytime temperature > summer nighttime temperature. The ranking of the maximum temperature difference was summer daytime temperature (4.68 °C) > winter nighttime temperature (4.6 °C) > winter daytime temperature (3.6 °C) > summer nighttime temperature (2.37 °C).

3.2. Correlation between air temperature and urban morphology parameters

The proposed empirical model was based on the comprehensive outdoor temperature [30]. The empirical prediction model used the T_{min} , T_{avg} , and T_{max} to predict the daytime, nighttime, and average temperatures. Table 4 shows the correlation coefficient between the urban morphology parameters and the temperature in the different buffer zones. The morphology parameters had the highest correlation with the temperature in the buffer with the 100 m radius; therefore, 100 m was selected as the radius in the model.

Table 5 shows the relationship between the morphological parameters and the temperature change and the results of the sensitivity analysis. Not all the parameters had an effect on the temperature. The *RH* and *SVF* were not significantly correlated with T_{min} in winter and summer. The *RH* was not significantly correlated with T_{avg} in winter. The *RH* and *GnPR* were not significantly correlated with T_{max} in winter and summer. *WIND* was not significantly correlated with T_{max} in winter.

We divided the reference temperature into temperatures higher and

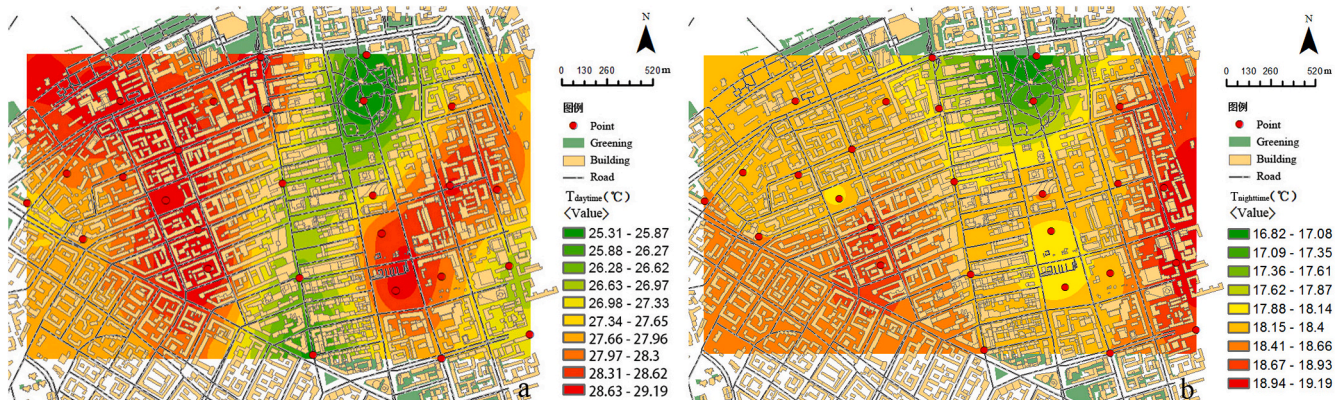


Fig. 5. The temperature distribution in summer (a) 14 : 00, June 13, 2018 (b) 2 : 00, June 13, 2018

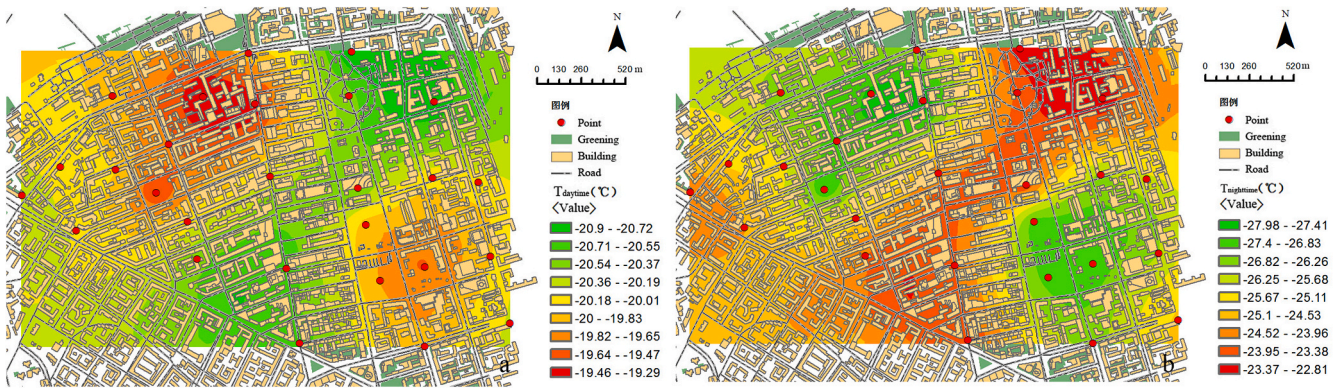


Fig. 6. The temperature distribution in winter (a) 14 : 00, Dec 21, 2017 (b) 2 : 00, Dec 21, 2017

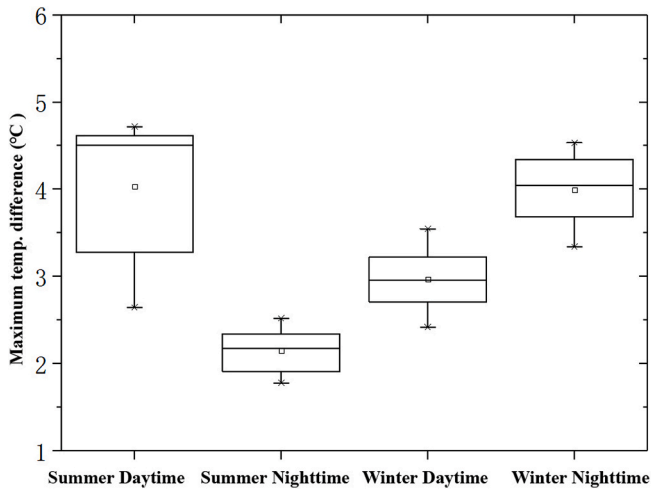


Fig. 7. Maximum temperature difference of each test point

lower than 0 °C so that the model is widely applicable. Table 6 shows the

Table 4

Correlation analysis of urban morphology parameters and temperature in different buffer radii ($P < 0.05$).

R^2	20 m		50 m		100 m		150 m		200 m	
	$T > 0\text{ }^{\circ}\text{C}$	$T < 0\text{ }^{\circ}\text{C}$	$T > 0\text{ }^{\circ}\text{C}$	$T < 0\text{ }^{\circ}\text{C}$	$T > 0\text{ }^{\circ}\text{C}$	$T < 0\text{ }^{\circ}\text{C}$	$T > 0\text{ }^{\circ}\text{C}$	$T < 0\text{ }^{\circ}\text{C}$	$T > 0\text{ }^{\circ}\text{C}$	$T < 0\text{ }^{\circ}\text{C}$
T_{min}	0.58	0.66	0.77	0.77	0.95	0.79	0.78	0.82	0.65	0.71
T_{avg}	0.41	0.76	0.89	0.90	0.92	0.90	0.83	0.85	0.67	0.73
T_{max}	0.33	0.81	0.63	0.81	0.90	0.88	0.81	0.81	0.78	0.79

Table 5

Correlation analysis of urban morphology parameters and temperature in 100 m radius.

Ref T >0 °C / <0 °C	T_{min}			T_{avg}			T_{max}		
	B	Sig.	P	B	Sig.	P	B	Sig.	P
Ref T	0.736/0.837	0.0/0.0	0.90/0.93	0.99/0.99	0.0/0.0	0.71/0.65	0.72/0.73	0.0/0.0	0.61/0.79
Solar	—	—	—	4.8E-05/5.7E-05	0.0/0.0	0.34/0.21	0.002/0.093	0.0/0.0	0.11/0.25
RH	0.001/0.002	0.7/0.2	0.02/0.02	-0.02/0.002	0.0/0.4	-0.07/0.21	0.001/0.001	0.09/0.06	0.1/0.06
WIND	-0.122/-0.09	0.0/0.0	-0.21/-0.17	-0.18/-0.021	0.0/0.0	-0.22/0.12	-0.33/-0.13	0.0/0.2	0.28/0.33
PAVE	0.02/0.004	0.0/0.0	0.31/0.08	0.01/0.007	0.0/0.0	0.33/0.07	0.015/0.001	0.0/0.0	0.21/0.07
GnPR	-0.31/-0.03	0.0/0.0	-0.21/-0.07	-0.04/0.06	0.0/0.0	-0.29/0.11	0.007/0.09	0.11/0.56	-0.28/-0.41
HBDG	-0.04/0.992	0.0/0.0	-0.21/0.38	-0.02/0.09	0.0/0.0	-0.09/0.06	-0.026/0.016	0.0/0.0	-0.07/0.08
WALL	2.2E-06/5.1E-06	0.0/0.0	0.1/0.6	2.1E-05/2.3E-05	0.0/0.0	0.32/0.21	2.2E-05/6.7E-06	0.0/0.0	0.11/0.28
SVF	-0.21/0.35	0.1/0.5	-0.3/0.5	0.81/1.32	0.0/0.0	0.07/0.12	1.43/1.41	0.0/0.0	0.31/0.39
Constant	3.021/3.261			3.177/3.434			1.542/1.228		
F	2907.12/2194.45			2280.11/2243.54			2242.45/2113.33		
R^2	0.96/0.90			0.97/0.95			0.96/0.91		
Std. Error	0.24/0.31			0.21/0.23			0.25/0.34		

Table 6

Correlation between air temperature and urban morphology parameters.

Ref $T > 0$ °C	$T_{min} = 3.021 + 0.736 \text{ Ref } T_{min} - 0.122WIND + 0.02PAVE - 0.31GnPR - 0.04HBDG + 2.2 \times 10^{-6}WALL (R^2 = 0.96, F = 2907.12, \text{Std. Error} = 0.24, \text{Sig.} = 0.00)$
	$T_{avg} = 3.177 + 0.997\text{Ref } T_{avg} - 0.02RH - 0.18WIND + 4.799 \times 10^{-5}SOLAR + 0.01PAVE - 0.04GnPR - 0.02HBDG + 2.121 \times 10^{-5}WALL + 0.81SVF (R^2 = 0.97, F = 2280.11, \text{Std. Error} = 0.21, \text{Sig.} = 0.00)$
	$T_{max} = 1.542 + 0.72\text{Ref } T_{max} + 0.002SOLAR + 0.01PAVE - 0.33WIND - 0.026HBDG + 2.2 \times 10^{-5}WALL + 1.43SVF (R^2 = 0.96, F = 2242.45, \text{Std. Error} = 0.25, \text{Sig.} = 0.00)$
Ref $T < 0$ °C	$T_{min} = 3.261 + 0.837\text{Ref } T_{min} - 0.09WIND + 0.004PAVE - 0.03GnPR + 0.992HBDG + 5.121 \times 10^{-6}WALL (R^2 = 0.90, F = 2194.45, \text{Std. Error} = 0.31, \text{Sig.} = 0.00)$
	$T_{avg} = 3.434 + 0.991\text{Ref } T_{avg} - 0.021WIND + 5.712 \times 10^{-5}SOLAR + 0.007PAVE + 0.06GnPR + 0.095HBDG + 2.33 \times 10^{-5}WALL + 1.32SVF (R^2 = 0.95, F = 2243.54, \text{Std. Error} = 0.23, \text{Sig.} = 0.00)$
	$sss (R^2 = 0.91, F = 2113.33, \text{Std. Error} = 0.34, \text{Sig.} = 0.00)$

$$NRMSE = \frac{RMSE_i}{RMSE_{i=ref}} = \sqrt{\frac{\sum_{i=1}^{Nd} \sum_{j=1}^{Ns} (T_{mea,i}(j) - T_{est,i}(j))^2}{\sum_{i=1}^{Nd} \sum_{j=1}^{Ns} (T_{mea,i}(j) - T_{ref,i})^2}} \quad (5)$$

where $NRMSE$ represents the standard root mean square error; $RMSE_i$ stands for the forecast temperature calculated by the equation in Table 6; $RMSE_{i=ref}$ is the reference temperature; $T_{mea,i}(j)$ is the measured temperature at point j on day i . The range of i is $1, 2 \dots Nd$; Nd represents the number of days used for verification, it is 5 this time. The range of j is $1, 2 \dots Ns$; Ns is the number of measurement points used for verification, it is 27 this time; $T_{est,i}(j)$ is the predicted temperature of measuring point j on day i ; $T_{ref,i}$ is the reference temperature of the city on day i . The error analysis results are shown in Fig. 8.

a. The reference temperature is above 0 °C: As Fig. 8a shows, the error range within ± 1 °C accounts for 96% of the prediction results (the light gray part in the figure), and the error range within ± 0.5 °C accounts for 54% of the prediction results (the dark gray part in the figure).

b. The reference temperature is below 0 °C: As Fig. 8b shows, the error range within ± 1 °C accounts for 85% of the prediction results (the light gray part), and the error range within ± 0.5 °C accounts for 41% of the prediction results (the dark gray part).

3.3. Multi-objective optimization of microclimate

The calculation took 180 h, and 42 generations and 5368 cases of the layout were calculated. The convergence occurred in the 30th generation, and the convergence correlation coefficient was 0.99. A total of 75 cases were obtained after convergence. The calculation results are shown in Fig. 9.

a. The optimal solution sets in different generations. As Fig. 9a shows, the number of the optimal solution set decreases with an increase in the optimization generation and stabilizes after convergence (30th generation), the final optimal solution set were 75 cases.

b. Schematic diagram of multi-axis space. As Fig. 9b shows, the sunshine duration, solar radiation, and wind speed influence each other (the closer the distance is to the origin of the coordinate axis, the better the results meet the optimization goal). Therefore, the proposed threshold is important. The black curve shows the change in the optimal solution set.

The curves of the three parameters are located in the middle region (middle of the coordinate axis). The optimization result shows that the performance was best for the wind speed optimization (closest to the origin of the coordinate axis).

c. Spatial distribution of Delaunay grids. d. Pareto frontier distribution map. As Fig. 9c and d shows, the final results were all located on the Pareto frontier, and the red point in the figure is the optimal solution set.

The influence of the morphological parameters on the microclimate can be determined by normalizing the difference between the optimization objective and the control variables of the simulation results. The results are shown in Fig. 10.

a. Optimal target value. As Fig. 10a shows, the optimization logic of the target was as follows: the optimization process starts by determining the sunshine duration and wind speed that result in the optimal temperature (the radiation temperature variance is the minimum). The wind speed is increased gradually to obtain the minimum wind speed difference. The maximum sunshine duration was determined based on the sunshine duration range. The solution conforming to the Pareto frontier is selected as part of the optimal solution set.

b. Morphology parameters value. As Fig. 10b shows, the following trends of the morphology parameters were observed. The building height increased, and the number of rows and columns decreased. The distance between the front and back buildings increased, and the distance between the left and right buildings decreased.

As shown in Table 7, as the number of generations increases, the building height and spacing gradually increase from south to north. The sunshine duration in the streets increased significantly (the blue color indicates insufficient sunshine duration), and the high-rise building areas meet the requirements for the sunshine duration. By the 30th generation, the optimum sunshine duration was obtained. The temperature changes in the block are balanced, and the wind speed also decreased significantly (the areas in red have higher wind pressure; the large the number of areas in red, the lower the wind speed is).

Due to a change in the optimization logic, in the initial generations, the building spacing and height changed based on the standard layout. In the middle of the optimization process, the individual building area and the height changed. After the convergence, the building density and block use efficiency were higher than before the convergence; however,

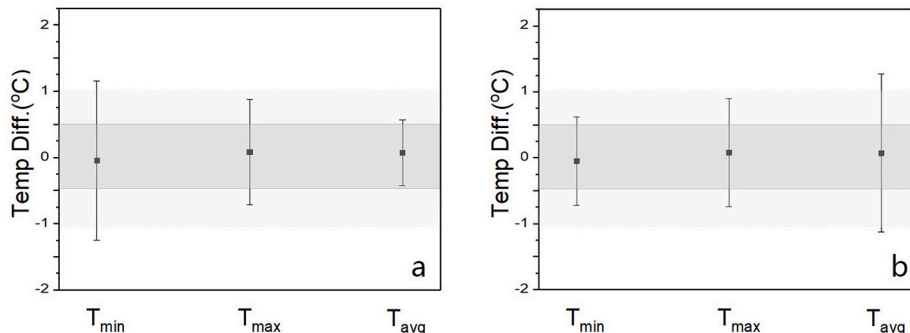


Fig. 8. Error analysis (a) the reference temperature is above 0 °C (b) the reference temperature is below 0 °C

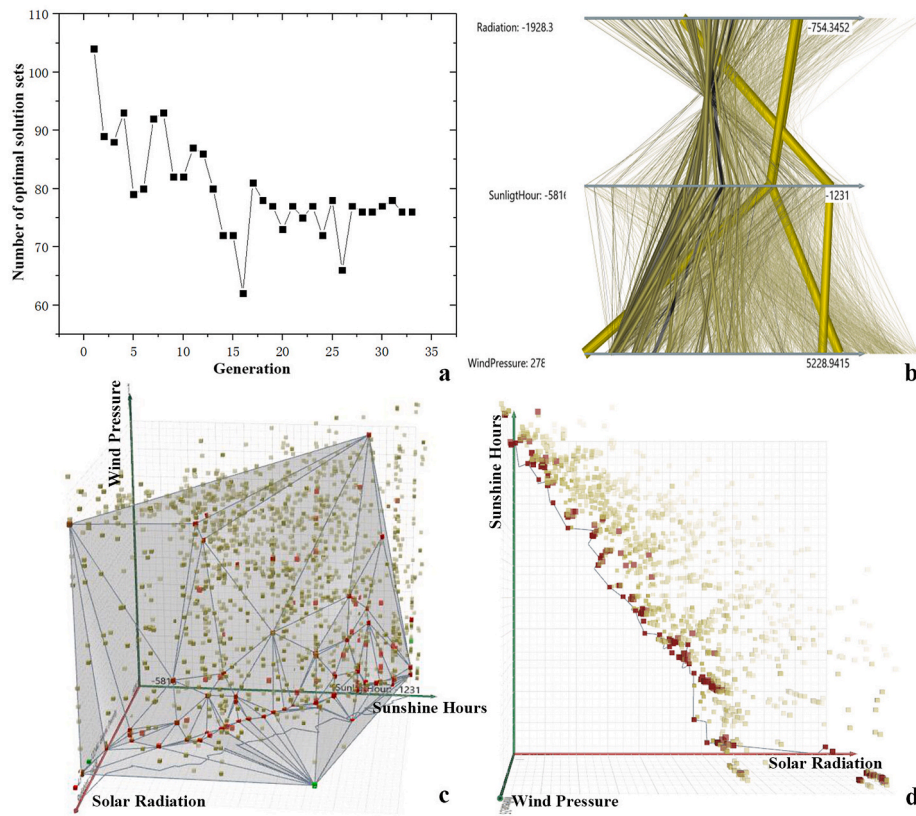


Fig. 9. Schematic diagram of calculation results. (a) The distribution of optimal solution sets in different generation (b) Schematic diagram of multi-axis space. (c) Delaunay grid spatial distribution. (d) Pareto frontier distribution map

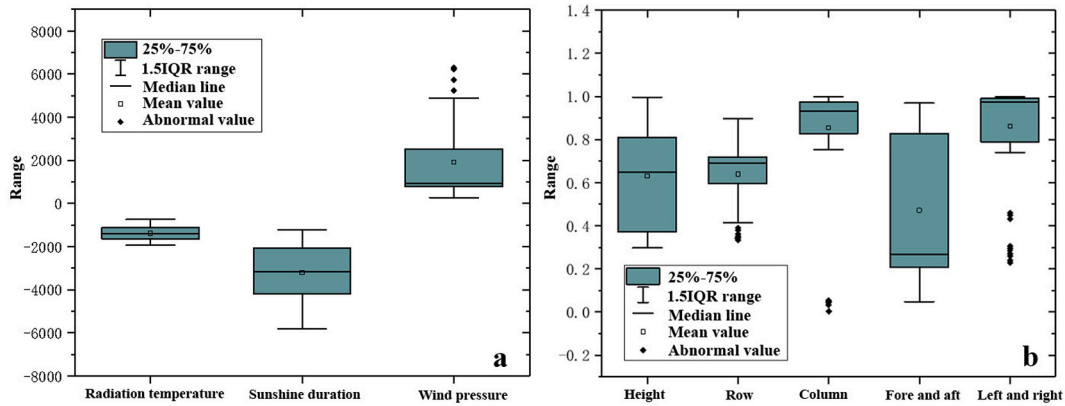


Fig. 10. Normalized differential analysis diagram (a) Optimal target value (b) Limit parameter value

the building height increased to meet the requirements of the sunshine duration and wind speed. Therefore, by considering the quantitative relationship between the calculated values (building height, number of buildings in the plot, number of buildings in the columns, and building spacing) and the mean value of the optimization target (sunshine hours, wind speed, solar radiation), a suitable optimization strategy of the urban microclimate can be determined.

4. Discussion and urban design recommendations

4.1. Parametric study

A parametric study was conducted to develop design recommendations on the street scale. Our results showed that the morphological

parameters that affected the outdoor temperature were *HBDG*, *SVF*, *GnPR*, and *PAVE*, which were simplified into building height, street width, and vegetation area. The height of the building and the street width influenced the *SVF*, *HBDG*, and *PAVE*. The calculated parameters of the simplified model are shown in Table 8.

Fig. 11 shows the calculation results of the street layout of Model 1 in the summer. As Fig. 11a shows, the T_{max} of all street widths decreases with an increase in the building height. For every 10 m increase in the building height, T_{max} decreases by 0.06 °C. At the same building height, for every 10 m increase in the street width, T_{max} increases by 0.17 °C.

As Fig. 11b shows, T_{min} of all street widths increases with the height of the buildings. T_{min} increases by 0.07 °C, on average, for every 10 m increase in building height. The street width has no significant influence on the temperature. In the high-rise areas, a 10 m increase in the street

Table 7

The calculation results of the optimal solution sets of different generation. (The image in the table is one of the optimal solution set in this generation)

	The optimization layout	Sunshine duration calculate	Wind pressure differential calculate	Temperature calculate
1st generation $R^2 = 0.43$				
5th generation $R^2 = 0.58$				
15th generation $R^2 = 0.65$				
30th generation $R^2 = 0.99$				

Table 8

Simplify model parameter Settings.

Model		1	2
Morphological parameter	Building levels (3 m per storey)	1–40	1–40
	Pavement width (W)	15 m/25 m/35 m/45 m	35 m
	Greenery strips width	5 m	5 m
	GnPR	1.5	0.5/1/1.5/2/2.5
Weather parameter		Summer	Winter
	SOLAR(W/m ²)	811	454
	Ref T_{avg} (°C)	22.8	−8.6
	Ref T_{max} (°C)	29.5	−3.3
	Ref T_{min} (°C)	22.1	−15.6

width results in an increase in the temperature of 0.01 °C.

As Fig. 11c shows, T_{avg} varies significantly in the building area with 2–20 floors; however, when the building has more than 20 floors, the temperature does not change significantly. The temperature of the narrow streets is significantly higher than that of the wide streets for the same building height. For buildings exceeding 30 floors (90 m), the street width has no effect on the T_{avg} .

Fig. 12 shows the calculation results of the street layout of Model 1 in the winter. As Fig. 12a shows, for building with 2–20 floors, T_{avg} increases significantly with an increase in the number of floors. For buildings with less than 20 floors, the change in the street width has a negligible influence on the temperature; For buildings with more than 20 floors, the temperature is higher for narrow streets, and the

temperature decreases by 0.05 °C for every 10 m increase in the street width.

As Fig. 12b shows, T_{min} increases by 1.3 °C for an increase in 10 floors (30 m) for all street widths. For buildings with less than 20 floors and street widths of 25 m, 35 m, and 45 m, the change in T_{min} is relatively small. For buildings with more than 20 floors, T_{min} decreases by 0.2 °C for every 10 m increase in the street width.

Fig. 13 shows the results of the street layout of Model 2. As Fig. 13a shows, T_{avg} has a linear relationship with the number of building floors in the summer as a result of the influence of GnPR. T_{avg} decreases by 0.5 °C for every 0.5 increase in the GnPR. A 0.5 increase in the GnPR results in a decrease in T_{min} of 0.8 °C (Fig. 13b). As Fig. 13c shows, for the same GnPR, T_{min} decreases by 0.3 °C for an increase in 10 m of the building height in the winter. At the same building height, T_{min} increases by 1.3 °C for every 0.5 increase in the GnPR.

Therefore, the GnPR should be increased on both sides of the street, and the height of the buildings should be minimized to achieve heating up in winter and cooling down in summer.

4.2. Urban design recommendations

This section focuses on the urban design recommendations at the block scale. Of the 5368 cases calculated in this study, the solution set for the microclimate optimization contains 75 optimal layouts. The optimal solution set does not provide a single index but rather a comprehensive index. The overall comfort of the people in a block should be comprehensively evaluated by using a block planning index, namely the BDG and HBDG.

Fig. 14 shows the results of the correlation analysis between BDG and

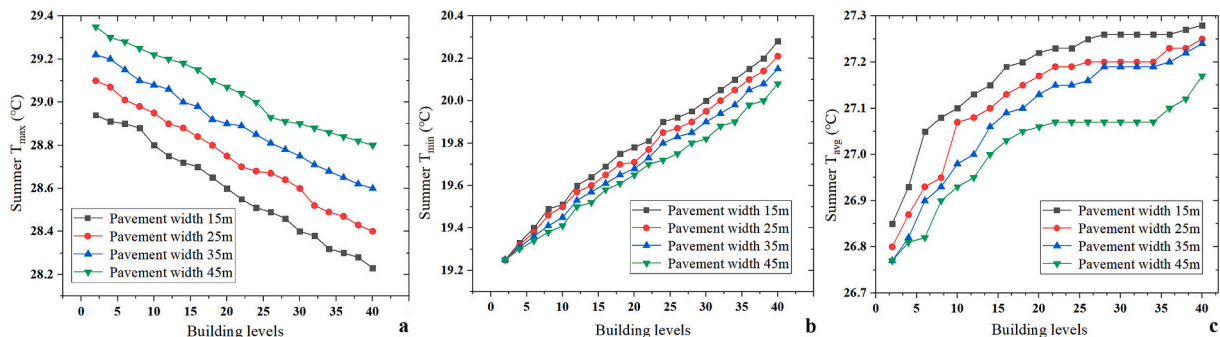


Fig. 11. Variation of temperature with number of building levels and pavement width in summer (a) T_{max} (b) T_{avg} (c) T_{min}

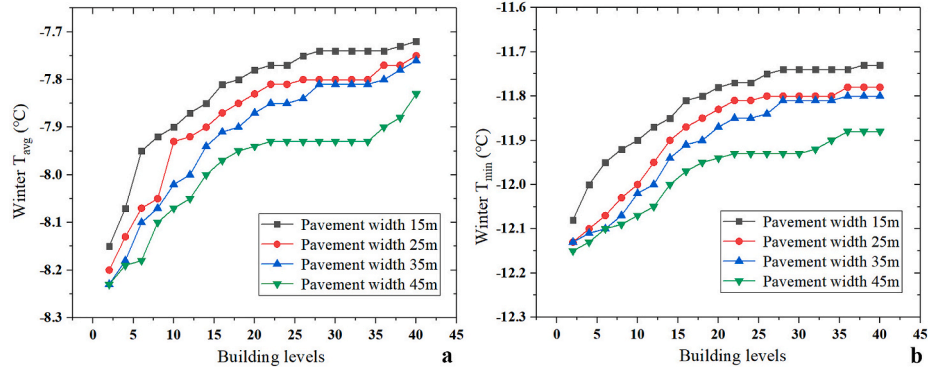


Fig. 12. Variation of temperature with number of building levels and pavement width in winter (a) T_{avg} (b) T_{min}

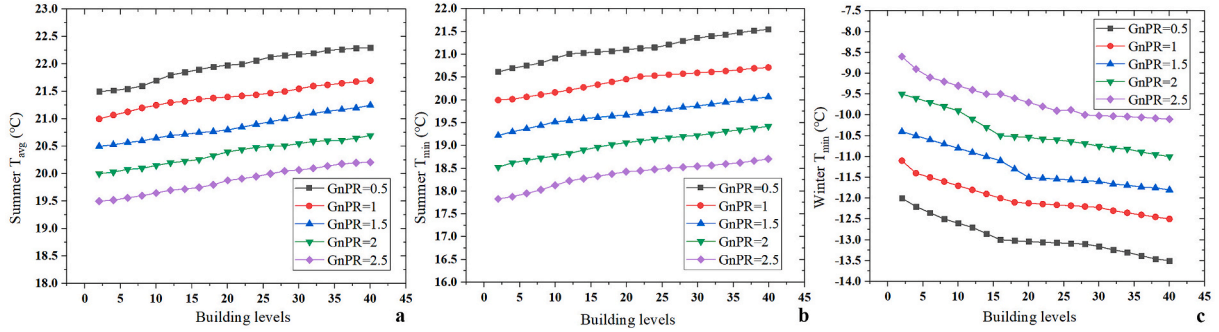


Fig. 13. Variation of temperature with number of building levels and GnPR (a) T_{max} in summer (b) T_{avg} in summer (c) T_{min} in winter

the proportions of the wind and thermal comfort zones. As shown in Table 7, the comfort zone for solar radiation is the orange and red area; the comfort zone for the wind speed is defined as the percentage of the area in red and orange in the whole area.

As Fig. 14a shows, BDG is negatively correlated with the proportion of thermal comfort zone ($R^2 = 0.57$), indicating that the area of the thermal comfort zone decreases as the building density increases.

Fig. 14b shows a significant positive correlation between the BDG and the proportion of the comfort zone for wind speed ($R^2 = 0.88$). This result demonstrates that the area of the wind comfort zone increases significantly as BDG increases. The relationship between BDG and the wind speed and solar radiation is the opposite. Therefore, there is a range where the comfort zones for both the wind speed and solar radiation coincide. Finding a compromise set of two fitted curves, the range for BDG is 20%–30%.

Fig. 15 shows the results of the correlation analysis between $HBDG$ and the proportions of the wind speed and thermal comfort zones. As Fig. 15a shows, $HBDG$ is negatively correlated with the proportion of

thermal comfort zone ($R^2 = 0.66$). This R^2 value is higher than that of BDG . It is worth noting that the overall dispersion degree in the fitting diagram is low, and only two points have high dispersion degrees. The R^2 is 0.85 if these cases are excluded, indicating that the primary factor influencing the proportion of the thermal comfort zone is the $HBDG$.

As Fig. 15b shows, $HBDG$ is positively correlated with the proportion of the wind speed comfort zone ($R^2 = 0.64$). An increase in $HBDG$ significantly reduces the wind speed in the winter and increases the proportion of the wind speed comfort zone. Since the wind speed and thermal comfort zones in the block are relatively balanced, when finding a compromise set of two fitted curves, the $HBDG$ should range from 3 to 5.

5. Conclusion

In this study, empirical models were developed to predict spatial and temporal changes in the air temperature at the street scale. A parametric study was conducted to assess the impact of urban morphology on the

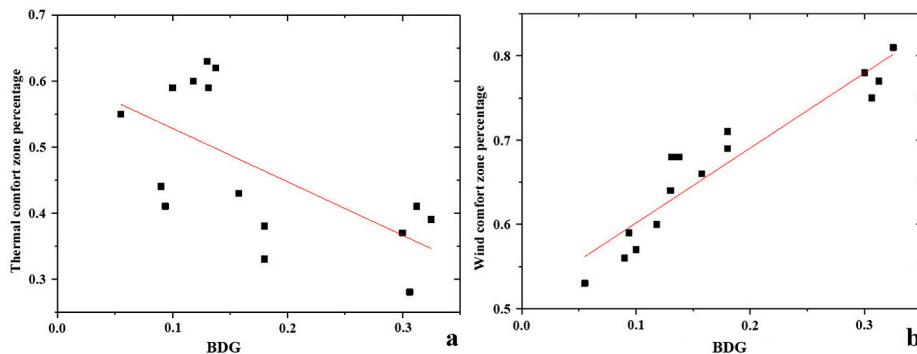


Fig. 14. Correlation of BDG with (a) thermal and (b) wind comfort zone percentage

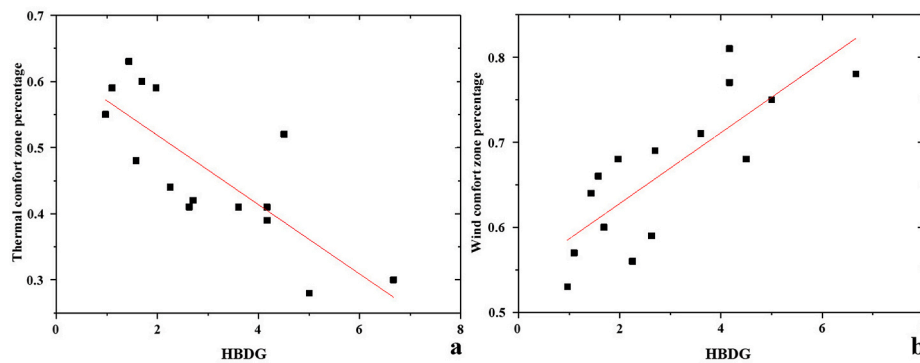


Fig. 15. Correlation of HBDG with (a) thermal and (b) wind comfort zone percentage

air temperature in summer and winter. It was found that an increase in the number of building levels and a reduction in the street width in the summer reduced T_{max} and T_{avg} and increased T_{min} . An increase in the $GnPR$ of 0.5 decreased the T_{avg} and T_{min} by 0.5 °C and 0.8 °C, respectively in the summer. In the winter, the temperature change is influenced by the number of building floors on both sides of the street. When the building floors are below 20, an increase in the building height and street width of 10 m, T_{avg} increases 0.03 °C and decreases 0.05 °C respectively. When the building floors are above 20, an increase in the building height and street width of 10 m, T_{min} increases by 0.4 °C and decreases by 0.05 °C respectively. At the same building height, T_{min} increases by 1.3 °C for every 0.5 increase in the $GnPR$.

A comprehensive microclimate optimization framework for urban design in cold regions in China was developed. The building height for a block should be regulated to avoid excessive differences in the building height. For the optimization of both the wind and thermal environment, the BDG had better not exceed 30%, and the $HBDG$ should be in the range of 3–5 for block unit (in Harbin is 400 m × 400 m). In addition, the proportion of open space in the block should be maximized on the premise of meeting the BDG and $HBDG$ index.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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